UNIT II: Data Analytics, AI and Automation in Manufacturing

Subject: 23A03703d - Smart Manufacturing

1. Role of Big Data in Manufacturing

1.1 What is big data?

Big data refers to data sets that are too large, too fast or too varied for traditional tools (such as a single spreadsheet or a simple database) to handle. In a factory, data comes from machines, PLCs, sensors, cameras, MES, ERP and quality systems.

1.2 The 5 V's of big data

- Volume: the amount of data (terabytes of sensor logs per month).
- Velocity: the speed at which data arrives (a vibration sensor may send thousands of samples per second).
- Variety: many formats (numbers, text, images, video, alarms).
- Veracity: how accurate and trustworthy the data is.
- Value: the useful insight that can be extracted and turned into money or savings.

KEY: Data by itself is not valuable; it becomes valuable only when it leads to a better decision.

1.3 Uses of big data in manufacturing

- Finding the root cause of defects by comparing thousands of production runs.
- Optimising machine settings (speed, temperature, pressure) for the best yield.
- Forecasting demand and planning inventory.
- Tracking energy use per product.
- Improving supplier quality through shared data.

1.4 Data architecture

- Data sources: machines, sensors, MES, ERP, quality and maintenance records.
- Data ingestion: collecting data through MQTT, OPC-UA or APIs.
- Data storage: data lakes (raw data), data warehouses (structured data), time-series databases (sensor readings).
- Processing: batch processing (for historical reports) and stream processing (for live alerts) with tools such as Apache Spark or Kafka.
- Analytics: statistics, machine learning and visualisation dashboards.

1.5 Data quality issues

- Missing values when a sensor fails or a network drops.
- Noise and outliers from electrical interference.
- Inconsistent timestamps between machines (clock drift).
- Labels that are wrong or missing (for example, defects not recorded correctly).

KEY: Clean data is the foundation: a model trained on poor data will give poor results.

Exam Q: Explain the 5 V's of big data with a manufacturing example for each.

2. Machine Learning and Deep Learning for Smart Manufa

2.1 Machine learning (ML)

Machine learning lets a computer learn patterns from data instead of following fixed rules written by a programmer.

- Supervised learning: learns from labelled examples. Regression predicts a number (for example, the remaining life of a bearing). Classification predicts a category (for example, good part or defective part).
- Unsupervised learning: finds structure without labels. Clustering groups similar production batches or machine behaviours.

- Reinforcement learning: learns by trial and reward. Used to tune robot control or scheduling policies.

2.2 Common ML algorithms in factories

- Linear and logistic regression: simple, explainable baselines.
- Decision trees and random forests: good for mixed data and for showing which factors matter.
- Support vector machines: classification with small to medium data sets.
- k-means clustering: grouping operating modes of a machine.
- Gradient boosting: strong performance on tabular process data.

2.3 Deep learning (DL)

Deep learning uses neural networks with many layers. It works very well on images, sound and sequences.

- Convolutional neural networks (CNN): inspect camera images to detect scratches, cracks or missing components.
- Recurrent networks and LSTMs: model time-series sensor data, such as temperature or current over time.
- Autoencoders: learn to rebuild normal data; large rebuild error signals an anomaly.
- Transformers: increasingly used for log analysis and combining several data types.

2.4 The ML workflow

1. Define the business problem (for example, reduce motor failures by 30 percent).
2. Collect and label data.
3. Clean and engineer features (for example, average vibration per minute, peak current).
4. Split data into training, validation and test sets.
5. Train and tune models.
6. Evaluate using suitable metrics (accuracy, precision, recall, mean absolute error).
7. Deploy to the edge or cloud and monitor for drift.

8. Retrain when the process changes.

2.5 Metrics to remember

- Precision: of the items flagged as defective, how many really are.
- Recall: of all real defects, how many were flagged.
- False alarms and missed failures have different costs; choose the metric that matches the cost.

Exam Q: Why is recall more important than precision for detecting a critical equipment failure?

3. Predictive Maintenance and Anomaly Detection

3.1 Maintenance strategies

Strategy	When work is done	Drawback
Reactive (run to failure)	After breakdown	Long downtime, damage to other parts
Preventive	At fixed intervals	Parts replaced even when still good
Condition-based	When a measured condition crosses a limit	Needs sensors and thresholds
Predictive	Before failure, based on forecast	Needs data, models and skill

- Predictive maintenance uses models to estimate the remaining useful life (RUL) of a component and schedule work at the best time.

3.2 Common condition indicators

- Vibration: rising amplitude or new frequency peaks indicate bearing or gear wear.
- Temperature: a slow rise can show lubrication problems.
- Motor current signature: changes reveal rotor or load problems.
- Acoustic emission: detects cracks and leaks.

- Oil analysis: metal particles show wear.

3.3 Anomaly detection

An anomaly is a pattern that does not match normal behaviour. Methods include:

- Statistical limits: flag values beyond mean plus or minus three standard deviations.
- Moving averages and control charts (Shewhart, CUSUM) for slow drifts.
- Isolation forest: isolates unusual points quickly in multi-sensor data.
- One-class SVM: learns the boundary of normal data.
- Autoencoder reconstruction error: a high error means the pattern is unfamiliar.

KEY: Anomaly detection tells you that something is unusual; diagnosis tells you what is wrong.

3.4 Workflow of a predictive maintenance system

- Collect vibration, temperature and current data at the edge.
- Extract features in the time and frequency domains (RMS, kurtosis, FFT peaks).
- Score each window with the model; compare with thresholds.
- Raise an alert and create a work order in the CMMS.
- Record the outcome to improve the model later.

3.5 Benefits

- Less unplanned downtime and fewer emergency repairs.
- Lower spare-part inventory because parts are ordered in time.
- Longer machine life and better safety.

4. Digital Twin Technology

4.1 Definition

A digital twin is a virtual representation of a physical asset, process or system that is kept up to date with live data from the real one. It can be used to monitor, simulate, predict and optimise.

KEY: The three parts of a digital twin: the physical object, its virtual model, and the two-way data connection between them.

4.2 Types of digital twin

- Component twin: a bearing or a motor.
- Asset twin: a complete machine such as a CNC or a robot.
- System twin: a production line or a cell.
- Process or plant twin: the whole factory, including material flow and energy.

4.3 Levels of twin maturity

- Digital model: no automatic data link; a static design model.
- Digital shadow: one-way data flow from physical to virtual.
- Digital twin: two-way link; the virtual model can also send commands or changes back.

4.4 What a digital twin enables

- Virtual commissioning: test PLC programs against the virtual line before the real one is built.
- What-if analysis: check the effect of a faster speed without risking real equipment.
- Remote monitoring and diagnosis.
- Predictive maintenance using the twin's physics-based model together with data-based ML.
- Training operators in a safe virtual environment.

4.5 Building blocks

- Geometric model: CAD and 3D shape.
- Physics model: equations for heat, flow, stress and motion.
- Behaviour model: rules or learned models of how the system responds.
- Data integration: IIoT connectivity and time-series storage.
- Visualisation: 3D dashboards, AR overlays.

4.6 Example

A pump has a digital twin built from its CAD model and fluid equations. Live pressure and flow data update the twin. When the twin's predicted efficiency drops below the measured efficiency, the team inspects the impeller for wear before a leak occurs.

Exam Q: Differentiate between a digital model, a digital shadow and a digital twin.

5. Industrial Automation: PLCs, SCADA and DCS

5.1 Programmable Logic Controller (PLC)

A PLC is a rugged industrial computer that reads inputs, runs a control program and drives outputs in a repeating cycle called the scan cycle.

- Scan cycle: read inputs -> execute program -> update outputs -> handle communication -> repeat.
- Typical scan times are a few milliseconds.
- Programming standard IEC 61131-3 includes ladder diagram (LD), function block diagram (FBD), structured text (ST), sequential function chart (SFC) and instruction list.
- Used for discrete manufacturing: conveyors, packaging, assembly stations, machine tools.

5.2 Ladder logic example

A motor start-stop circuit: a START contact and a STOP contact control the motor coil. Pressing START latches the motor on through a holding contact; pressing STOP breaks the latch. In ladder notation, rungs are read left to right like rungs of a ladder, from power rail to power rail.

5.3 SCADA (Supervisory Control and Data Acquisition)

SCADA is a system that monitors and controls geographically spread or large processes from a central place.

- Components: sensors and PLCs or RTUs (remote terminal units) at the field, communication network, SCADA server, historian database, and HMI screens.
- Functions: data acquisition, supervisory control (setpoints, start and stop), alarms, trends, reports and logs.
- Used in water treatment, power distribution, oil pipelines, railways and large plants.

5.4 DCS (Distributed Control System)

A DCS is a control system in which controllers are distributed across the plant but managed in an integrated way.

- Designed mainly for continuous and batch process industries: chemicals, refineries, pharmaceuticals, power plants.
- Strong points: built-in redundancy, process control blocks (PID loops), tight integration of control and operations.
- Distinction: SCADA is strong for wide-area monitoring; DCS is strong for one process site with complex control loops.

Feature	PLC	SCADA	DCS
Main role	Local logic control	Supervisory monitoring	Integrated process control
Typical industry	Discrete manufacturing	Utilities, pipelines	Chemicals, refineries
Scale	Machine or cell	Wide area	Single large plant
Redundancy	Optional	Server level	Built in

Exam Q: Compare PLC, SCADA and DCS in a short table.

6. Autonomous Robots and Cobots

6.1 Industrial robots

An industrial robot is a programmable mechanical arm with several joints (axes). Typical types include articulated, SCARA, delta and Cartesian robots. They are used for welding, painting, pick-and-place, machine tending and assembly.

6.2 Autonomous mobile robots (AMRs) and AGVs

- Automated guided vehicles (AGVs) follow fixed paths such as magnetic tape or wires.
- Autonomous mobile robots (AMRs) use sensors, maps and planning to find their own route around people and obstacles.
- Used for moving materials between stations and warehouses.

6.3 Collaborative robots (cobots)

A cobot is designed to work safely alongside people without a safety cage.

- Features: force and torque sensing, rounded joints, speed and separation monitoring, power and force limiting.
- Safety standards: ISO 10218 for industrial robots and ISO/TS 15066 for collaborative operation.
- Advantages: easy to program by hand-guiding, quick to redeploy, low cost for small batches.
- Limitations: slower and lower payload than traditional industrial robots.

KEY: Cobots do not replace safety analysis; each application must still be risk-assessed.

6.4 Robot programming methods

- Teach pendant: operator moves the robot and records points.
- Hand guiding: operator moves the arm physically (common for cobots).
- Offline programming: program built in simulation and then loaded.
- Vision-guided picking: camera finds the part position and robot adapts.

Exam Q: State two differences between an AGV and an AMR.

7. Additive Manufacturing (3D Printing)

7.1 Basic idea

Additive manufacturing builds a part layer by layer from a digital 3D model, adding material only where it is needed. It is the opposite of subtractive machining, which removes material.

7.2 Main processes

- Fused deposition modelling (FDM / material extrusion): melted thermoplastic is extruded through a nozzle.
- Stereolithography (SLA / vat photopolymerisation): a UV laser cures liquid resin layer by layer.
- Selective laser sintering (SLS / powder bed fusion): a laser fuses nylon or other powder.
- Metal powder bed fusion (SLM, DMLS): a laser melts metal powder for functional metal parts.
- Binder jetting: a liquid binder joins powder layers; parts are then sintered.
- Directed energy deposition: metal is melted as it is deposited, used for repairs.

7.3 Advantages in smart manufacturing

- Complex geometry at no extra cost (internal channels, lattices).
- Small batches and customised products.
- Rapid prototyping and fewer tooling costs.
- Spare parts printed on demand near the point of use.
- Lightweight parts for aerospace and automotive.

7.4 Limitations

- Slower than mass moulding for large volumes.
- Surface finish and dimensional accuracy may need post-processing.
- Material properties and repeatability need careful control and inspection.

7.5 Digital thread

A digital thread connects the design file, print parameters, build records and inspection data for each part. This allows traceability and quality certification.

8. Case Studies of Smart Factories

8.1 Siemens Electronics Works Amberg (Germany)

- A highly automated plant producing automation controllers. Products are tracked through the line, and the plant is often cited as a reference for digital manufacturing.
- Features: extensive use of PLCs, MES, robotics and digital planning, with very high quality levels reported.
- Lessons: integrate IT and OT early; use the data from one stage to improve the next.

8.2 BMW Spartanburg and other automotive plants

- Automotive plants use robots, vision systems and AGVs/AMRs for flexible assembly.
- Lessons: flexible lines that can build several models; digital planning with simulation before construction.

8.3 General lessons from smart factory cases

- Start with a clear business goal such as less downtime or less scrap.
- Pilot one line, measure results, then scale to more lines.
- Invest in skills and change management as much as in technology.
- Measure benefits with agreed KPIs such as OEE, first-pass yield and energy per unit.

KEY: OEE (Overall Equipment Effectiveness) = Availability x Performance x Quality.

Exam Q: Write the steps you would follow to plan a predictive maintenance project in a plant.

9. Worked Examples and Revision

9.1 Worked example: bearing health monitoring

A motor bearing is monitored with an accelerometer sampling at 10 kHz.

- Step 1: every minute, compute the RMS vibration value of the last 60 seconds.
- Step 2: learn the normal range from six weeks of healthy data (for example, 0.8 to 1.2 mm/s).
- Step 3: set an alert at 1.8 mm/s and a shutdown limit at 2.5 mm/s.
- Step 4: use an autoencoder on frequency-domain features to catch new fault types that thresholds miss.
- Step 5: when the alert fires, the CMMS creates a work order; the technician replaces the bearing in the next planned stop.
- Result: the failure is caught before the motor overheats, avoiding a two-day line stop.

KEY: Always combine simple thresholds (fast and explainable) with advanced models (sensitive to new patterns).

9.2 Worked example: choosing a robot for a packaging cell

- Task: pick cartons weighing 3 kg and place them on a pallet, working next to a packer.
- Requirement: human workers enter the cell for refilling, so the robot must be a cobot.
- Check payload (at least 3 kg plus gripper weight), reach, speed and safety features.
- Decision: a cobot with force limiting and a safety-rated stop, programmed by hand guiding.

9.3 Worked example: PLC logic for a conveyor

- Inputs: START button, STOP button, photo sensor at the end of the belt.
- Output: conveyor motor.

- Logic: motor runs when START is pressed and STOP is not pressed; the motor stops automatically when the photo sensor sees a full box for 3 seconds.
- Test: simulate each input in the PLC software before connecting the motor.

9.4 Revision checklist

- Can you explain the 5 V's of big data?
- Can you name a supervised and an unsupervised learning method with an example?
- Can you describe the difference between anomaly detection and diagnosis?
- Can you draw the PLC scan cycle?
- Can you compare SCADA and DCS?
- Can you list two additive manufacturing processes and their uses?
- Can you define OEE?

9.5 Short answer practice

Exam Q: What is the role of an edge gateway in a predictive maintenance system?

Exam Q: Why do cobots need force and torque sensing?

Exam Q: Give two reasons why a digital twin is more useful than a static CAD model.